ASSESSMENT TOOL - 3

SIMATS ENGINEERING

COURSE : ARTIFICIAL INTELLIGENCE "C"

NAME : A. SWAGIATH

REG. NO : 192526005 6

ASSIGNMENT - 3

Types of Intelligent Agents in a Smart Home.

Introduction

An intelligent agent is a system that observes its environment and takes action to achieve a goal. In a smart home, it controls lights, temperature, and security automatically using sensors and user preferences.

There are four types of intelligent agents:

1. Simple Reflex Agent

A simple reflex agent takes action based on the current situation. It follows fixed rules and does not remember past event.

Examples:

* Turns on light when motion is detected
* Turns on the AC when the temp is high

Advantages:

* Easy to design.
* Responds quickly
* Uses less memory

Disadvantages:

* Cannot learn user preferences.
* Cannot remember past situations.

2. Model - Based Agent

A model based agent remembers information about its environment and uses it to make decisions.

Examples:-

* Remembers which rooms are occupied.
* Tracks temperature changes.
* Adjust lights based on room usage.

Advantages:

* Uses past information
* Makes betters decisions
* Works well in changing situations.

Disadvantages:

* Requires more memory
* Needs accurate information about the environment.

3. Goal - Based Agent.

A goal - based agent takes action to achieve a particular goal.

Examples :

* Turns off lights to save electricity
* Locks doors when people leave home.
* Adjusts the temperature to maintain comfort.

Advantages.

* Works toward specific goals.
* Can plan actions.
* Helps solve problems.

Disadvantages:

* Planning may take more time
* Needs clearly defined goals.

4. Utility - Based Agent.

A utility - based agent selects the best action by comparing different choice. It considers comfort, cost, energy and security.

Examples:

* Maintains a comfortable temperature while saving electricity.
* Adjusts lighting to reduce energy use.
* Balances comfort and electricity costs.

Advantages:

* Makes better choices
* Balances different needs
* Improves energy efficiency.

Disadvantages:

* More difficult to design.
* Requires more processing.

Conclusion:

Intelligent agent make smart homes more comfortable, safe and energy-efficient. A combined of different agents provides better result.

2.

Uninformed Search Strategies in an Unknown Maze

Introduction:

Uninformed search is a method used in artificial intelligence to find a solution without knowing which path is best.

1. Uniform Cost Search (UCS)

Uniform cost search selects the path with the lowest total cost first.

working:

1. Start from the initial position.
2. Check the available paths
3. Calculate the cost of each path.
4. Choose the path with the lowest cost.
5. Continue until the exit is found.

Example:

If one path cost 10 units and another costs 5 units., UCS choose the path costing 5 units first.

Advantages:

* Finds the lowest-cost path
* works with different path cost.
* Can find a solution when one exist
* under suitable conditions.

Disadvantages:

* Requires more memory.
* Can take more time for large mazes.
* May explore many unnecessary paths.

Complexity:

* Time: can be very high for large search spaces.
* Space: Requires more memory because it

2. Iterative Deepening search (IDS)

Iterative Deepening search searches the maze by gradually increasing the depth limit.

Working:

1. Start from the initial position
2. Search up to depth 0
3. If the exist is not found, increase the depth to 1.
4. Continue increasing the depth.
5. Stop when the exist is found

Example:

The robot first checks nearby positions, then searches farther positions until it finds the exit.

Advantages:

* Uses less memory
* Finds a solution if one exists under suitable conditions.
* Works when all movement have equal cost.

Disadvantages:

Repeats some searches

May take more time for deep mazes.

Complexity:

Time: $O(b^d)$

Space: $O(bd)$

3. Difference Between UCS and IDS

Feature	UCS	IDS
Working	Chooses the lowest cost path	Increase depth step by step.
Memory	Requires more memory	Requires less memory
Time	can be high	can be high
Best use	Different path cost	Equal path costs

4. Types of Intelligent Agents in maze Solving.

Simple Reflex agent.

Model - Based Agent

Goal - Based Agent

Utility - Based Agent.

Conclusion:-

UCS finds the lowest - cost path but may requir more memory. IDS uses less memory and searches by increasing the depth limit.